INTERTWINING OF THE GYSIN-AT-INFINITY CLASS WITH THE LOCALIZED P-TH CHERN CLASS UNDER THE FURTHER HYPOTHESIS THAT THE RANK OF THE QUOTIENT IS LESS THAN P

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The statement inherits the full setup of the same chapter result labelled lemma-localized-chern-pre (retained as a tagged navigation pointer) together with the additional rank hypothesis rank(Q|_T) < p; the arguments of the lemma are the two chain computations written out inside the counted span.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FAV (sources/stacks-chow/pinned-original-source.tex line 10353).

Allowed standard prerequisites (editorial).

- The localized Chern class setup and the class c'_p(Q). Same chapter, Lemma labelled lemma-localized-chern-pre (source line 10204), whose setup the statement inherits, invoked at statement line 10354 and proof line 10366. Fix the objects Q, Z, T, W, b and the classes c_p(Q), c'_p(Q) and the complement assumption rank(Q|_T) < p.
- The Gysin-at-infinity bivariant class C and its defining property. Same chapter, Lemma labelled lemma-gysin-at-infty (source line 10026), invoked at statement line 10357. Provides the class C with C cap alpha = i_infinity^* beta for a suitable beta, which is the third equality of the displayed chain.
- Two intermediate computations identifying the restricted classes. Same chapter, Lemmas labelled lemma-silly (source line 9695), lemma-silly-silly (source line 9792), lemma-homomorphism-pre (source line 10132) and lemma-bivariant-zero (source line 6585), invoked at proof lines 10368, 10395, 10400 and 10402. Supply the individual steps of the chain of eight equalities, namely the description of c'_p(Q) through E, the comparison on i_infinity^* beta, the commutation with i_infinity^* and the reduction to testing against cycle classes.

Source reference. Original lemma 0F9K: sources/stacks-chow/pinned-original-source.tex, source lines 10204–10204. Same-chapter pointer into the author environment lemma

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at source line 10204; invoked at proof line 10366 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02UC: sources/stacks-chow/pinned-original-source.tex, source lines 6585–6585. Same-chapter pointer into the author environment lemma at source line 6585; invoked at proof line 10368 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9H: sources/stacks-chow/pinned-original-source.tex, source lines 9695–9695. Same-chapter pointer into the author environment lemma at source line 9695; invoked at proof line 10395 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9I: sources/stacks-chow/pinned-original-source.tex, source lines 9792–9792. Same-chapter pointer into the author environment lemma at source line 9792; invoked at proof line 10400 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FAR: sources/stacks-chow/pinned-original-source.tex, source lines 10132–10132. Same-chapter pointer into the author environment lemma at source line 10132; invoked at proof line 10402 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Lemma 0F9K assume $Q|_T$ is isomorphic to a finite locally free $\mathcal{O}_T$-module of rank $< p$. Denote $C \in A^0(W_\infty \to X)$ the class of Lemma 0F9J. Then*

$$C \circ c_p(Q|_{X\times\{0\}}) = C \circ (Z \to X)_* \circ c'_p(Q) = c_p(Q|_{W_\infty}) \circ C$$

Proof. The first equality holds because $c_p(Q|_{X\times\{0\}}) = (Z \to X)_* \circ c'_p(Q)$ by Lemma 0F9K. We may prove the second equality one cycle class at a time (see Lemma 02UC). Since the construction of the bivariant classes in the lemma is compatible with base change, we may assume we have some $\alpha \in \mathrm{CH}_k(X)$ and we have to show that $C \cap (Z \to X)_*(c'_p(Q) \cap \alpha) = c_p(Q|_{W_\infty}) \cap C \cap \alpha$. Observe that

$$\begin{aligned}
C \cap (Z \to X)_*(c'_p(Q) \cap \alpha) &= C \cap (Z \to X)_*(E \to Z)_*(c'_p(Q|_E) \cap C \cap \alpha) \\
&= C \cap (W_\infty \to X)_*(E \to W_\infty)_*(c'_p(Q|_E) \cap C \cap \alpha) \\
&= C \cap (W_\infty \to X)_*(E \to W_\infty)_*(c'_p(Q|_E) \cap i_\infty^*\beta) \\
&= C \cap (W_\infty \to X)_*(c_p(Q|_{W_\infty}) \cap i_\infty^*\beta) \\
&= C \cap (W_\infty \to X)_* i_\infty^*(c_p(Q) \cap \beta) \\
&= i_\infty^*(c_p(Q) \cap \beta) \\
&= c_p(Q|_{W_\infty}) \cap i_\infty^*\beta \\
&= c_p(Q|_{W_\infty}) \cap C \cap \alpha
\end{aligned}$$

as desired. For the first equality we used that $c'_p(Q) = (E \to Z)_* \circ c'_p(Q|_E) \circ C$ where $E \subset W_\infty$ is the inverse image of Z and $c'_p(Q|_E)$ is the class constructed in Lemma 0F9H. The second equality is just the statement that $E \to Z \to X$ is equal to $E \to W_\infty \to X$. For the third equality we choose $\beta \in \mathrm{CH}_{k+1}(W)$ whose restriction to $b^{-1}(\mathbf{A}^1_X)$ is the flat pullback of α so that $C \cap \alpha = i_\infty^*\beta$ by construction. The fourth equality is Lemma 0F9I. The fifth equality is the fact that $c_p(Q)$ is a bivariant class and hence commutes with i_∞^*. The sixth equality is

Lemma 0FAR. The seventh uses again that $c_p(Q)$ is a bivariant class. The final holds as $C \cap \alpha = i_\infty^* \beta$. $\square$